EXPERIMENT - 12

Aim:

End - End Communication at Transport

i) Implement echo client server using TCP/UDP socket

ii) Implement a chat program using socket programming.

TCP :-

```
import socket
import threading

# - Server - -
def start_server():
    server = socket.socket(socket.AF_INET, socket.
    server.bind(("127.0.0.1", 5000))                STREAM)
    server.listen(1)
    print(f"[SERVER] Connected by {addr}")

while True:
    data = conn.recv(1024).decode()
    if not data or data.lower() == "quit":
        print("[SERVER] Connection closed by clie
        break
    print(f"[SERVER] Received: {data}")
    conn.send(f"Server received: {data}".e
conn.close()
server.close()

# -- CLIENT - -
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_
                                                STREAM
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")
```

```
while True:
    msg = socket.socket(socket.AF_INET
    msg = input("Enter message (or 'quit' to exit):")
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1024).decode()
    print(f"[SERVER RESPONSE] {response}")
client.close()

# -- MAIN --
if __name__ == '__main__':
    server_thread = threading.Thread(target=start_server, daemon=True)
    server_thread.start()

    start_client()
```

OUTPUT :-

```
[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to Server
[SERVER] Connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):
Hello
[SERVER] Received : HELLO
[SERVER] RESPONSE Server recieved : Hello
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client
```

UDP:-

```
import socket
import threading
# -- SERVER --
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped.")
# -- CLIENT --
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
```

```
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] Closing connections...") break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[CLIENT] Stopped.")

# Main --
if __name__ == "__main__":
    threading.Thread(target = udp_server, daemon=True).start() udp_client().
```

OUTPUT:

[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to Server
[SERVER] Connection by ('127.0.0.1', 53458) Enter message (or 'quit' to exit)

~~Hello~~ Hi
[SERVER] Receiver : ~~Hello~~ Hi
[SERVER] [RESPONSE] Server received : Hello.
Enter message (or 'quit' to exit): quit
[SERVER] Connection ~~closed by client~~. Stopped.

RESULT:

Thus, the program was successfully executed.